PRISM: A Post-Prediction Reliability Framework for Evidence-Guided Correction and Selective Abstention

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***Abstract*—A model may produce a prediction with high confidence and still get the outcome wrong. This is a problem in sequential applications, where the system evolves over time and the data observed during deployment may differ from what the model observed during training. To address this issue we propose PRISM (Prediction Reliability and Intelligent Self-Correction Mechanism), a framework that evaluates the reliability of a prediction after it has been generated. It first estimates the possibility of an error and then examines the predictions identified as potentially unreliable. For such cases, PRISM analyses the available signals, searches for similar situations in historical data and checks whether an alternative prediction is supported by the available evidence. Based on the result of this process, it can retain the original prediction, replace it with a verified correction or abstain when the evidence is not enough to make a reliable decision.**

We study PRISM by a match's 2nd innings prediction in the Indian Premier League (IPL). Ball by ball records are converted into over level observations to study how the state of a match changes over an innings. To preserve the temporal order of the data, earlier IPL seasons are used for training, IPL 2024 is reserved for validation, and IPL 2025–2026 are used for out of time testing. The documented dataset consists of 8,665 over level observations from 475 matches, with the reported test partition containing 2,568 observations from 142 matches. In one experimental record, Logistic Regression, Random Forest and HistGradientBoosting achieve accuracies of 65.2%, 68.4% and 70.2%, respectively. The corresponding PRISM results show a selective accuracy of 82.4% at 88.5% coverage with a Brier score of 0.126 and an expected calibration error (ECE) of 0.019.

***Index Terms*—reliable prediction, selective prediction, abstention, error detection, evidence retrieval, post prediction correction, distribution shift, cricket analytics.**

I. Introduction

Accuracy and log loss are commonly used to measure the performance of machine learning models. While these metrics help compare different approaches, they do not fully address an important question i.e how should a system respond when it comes across a prediction that may not be reliable? This question becomes important in sequential applications.

A model's confidence can provide useful information about it's predictions but prediction with a high confidence is not necessarily a correct one. Calibration methods try to make predicted probabilities more consistent with the frequencies observed. Temperature scaling, for example, is a commonly used post-hoc calibration method [1]. Selective classification uses a different approach by allowing a model to withhold its prediction when the estimated risk is too high [2], [14]. Measures like the Brier score also help assess the quality of predicted probabilities rather than considering accuracy alone [10], [11]. In addition, out of distribution detection and ensemble based uncertainty estimation have been used until now to identify unfamiliar inputs and quantify uncertainty in model outputs [3], [4]. Although these methods provide ways to identify potentially unreliable predictions, determining whether an incorrect prediction can be corrected using available evidence remains a separate challenge.

PRISM (Prediction Reliability and Intelligent Self-Correction Mechanism) is proposed to address this problem. It works after a base model has generated its prediction and examines whether that output should be trusted. The framework estimates the risk of error, investigates the signals associated with a potentially unreliable prediction and finds relevant examples from historical data. Before an alternative is accepted, it is passed through a verification stage to determine whether the available evidence supports the proposed change. The system don't attempt to correct every prediction it flags. When the evidence is not sufficient, it can abstain rather than introduce a correction that may be less reliable than the original output.

The main contributions of this study are:

- We propose PRISM, a framework that performs reliability assessment and intervention after the initial prediction has been generated.
- We introduce an error diagnosis and historical evidence retrieval process to identify possible alternatives for predictions considered unreliable.
- We use a verification phase that determines whether a

proposed correction should be accepted with the final decision allowing the system to retain, correct or abstain from a prediction.
- We evaluate this framework on 2nd innings IPL prediction using chronological match level splits, prediction coverage and correction related metrics.

II. Related Work

A. Calibration and Predictive Uncertainty

Calibration examines whether predicted probabilities agree with actual outcomes. Guo *et al.* introduced temperature scaling as a simple way to improve neural network calibration [1], while Lakshminarayanan *et al.* explored deep ensembles for estimating predictive uncertainty [4]. PRISM uses these ideas as reliability signals since confidence alone cannot guarantee a correct prediction.

B. Selective Prediction

Selective prediction enables a model to abstain when the estimated risk is high, leading to a trade off between coverage and prediction accuracy. Geifman and El-Yaniv explored this approach for deep neural networks [2]. PRISM extends this idea by retrieving historical examples and verifying a possible correction before accepting it.

C. Out-of-Distribution Detection

Out of distribution detection identifies inputs that differ from the training data. Hendrycks and Gimpel proposed maximum softmax probability as a simple detection baseline [3]. Although such methods help flag unfamiliar inputs, they do not determine whether a prediction should be corrected or rejected. PRISM solves this through evidence retrieval and verification.

III. Problem Formulation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ denote a dataset of sequential prediction states, where $x_i \in \mathcal{X}$ contains the information available at a particular instance and $y_i \in \{0, 1\}$ denotes the eventual outcome. A base model f_θ produces

$$\hat{p}_i = f_\theta(x_i), \qquad \hat{y}_i = \mathbb{I}[\hat{p}_i \geq 0.5]. \tag{1}$$

PRISM receives x_i, $\hat{p}_i$, and inference time signals. It returns a decision

$$d_i \in \{\text{ACCEPT}, \text{CORRECT}, \text{ABSTAIN}\}. \tag{2}$$

The final prediction is defined only when the system accepts the original prediction or an alternative:

$$\tilde{y}_i = \begin{cases} \hat{y}_i, & d_i = \text{ACCEPT}, \\ \hat{y}_{c,i}, & d_i = \text{CORRECT}, \\ \perp, & d_i = \text{ABSTAIN}, \end{cases} \tag{3}$$

where $\perp$ denotes abstention and $\hat{y}_{c,i}$ is a candidate correction. The evaluation reports both the quality of accepted predictions and the fraction of inputs for which a prediction is returned.

IV. PRISM Framework

A. Overview

PRISM is divided into 5 stages: Detect, Diagnose, Correct, Verify and Decide. The base predictor remains a separate component. Fig. 1 shows the workflow.

B. Detection

The first stage of PRISM examines the prediction made by the base model and estimates the probability of it being incorrect. For this purpose, the detector uses a set of signals ϕ_i, which includes prediction confidence, probability margin, uncertainty estimates, similarity scores and model disagreement. The estimated error risk is given by

$$r_i = g_\psi(\phi_i), \qquad r_i \in [0, 1], \tag{4}$$

where a higher value of r_i indicates a greater probability of error. A threshold τ_r is then used to decide whether the prediction requires further correction:

$$q_i = \mathbb{I}[r_i > \tau_r]. \tag{5}$$

To prevent the detector from learning overly optimistic error estimates, its training targets are generated using out of fold or held out predictions rather than predictions obtained directly from the training samples.

C. Diagnosis

Once a prediction is flagged, PRISM examines the corresponding input to understand the context in which the prediction was made. This stage forms a diagnostic representation using the input state, the original prediction and its estimated risk:

$$z_i = h_\omega(x_i, \hat{p}_i, r_i). \tag{6}$$

For the IPL application, this representation describes the current match situation and provides a basis for comparing it with previously observed matches.

D. Evidence Guided Correction

PRISM uses previously observed examples as a source of evidence when considering a possible correction. Its reference memory is defined as

$$\mathcal{M} = \{(u_j, y_j)\}_{j=1}^{M}, \tag{7}$$

where u_j represents a historical state and y_j denotes its eventual outcome. For a flagged prediction, the retrieval module searches this memory for the K states most similar to the current representation:

$$\mathcal{N}_K(z_i) = \text{KNN}(z_i, \mathcal{M}, K). \tag{8}$$

The experiments use $K = 50$ neighbours. These retrieved examples are then used to propose an alternative prediction:

$$\hat{y}_{c,i} = C_\omega\left(x_i, \hat{y}_i, \mathcal{N}_K(z_i)\right). \tag{9}$$

The retrieved evidence is used to support the correction rather than replace the base model directly. To maintain a valid evaluation, the reference memory contains only examples from the training period.

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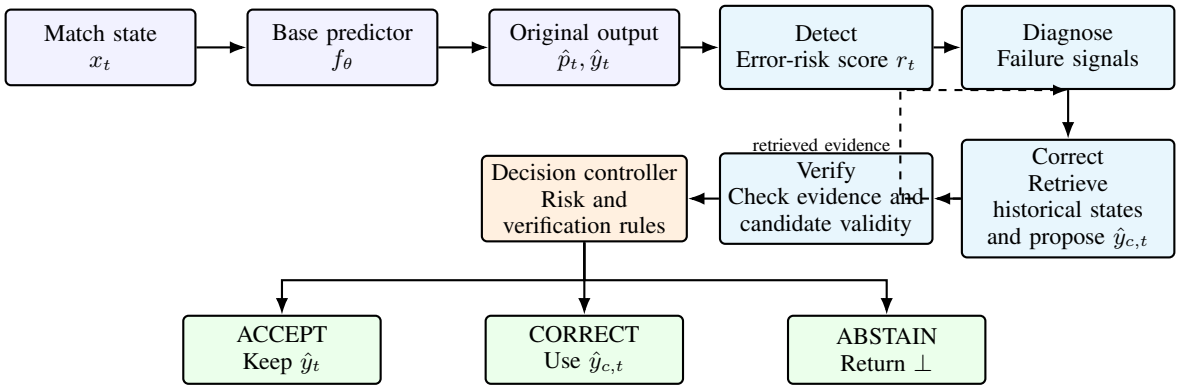


Fig. 1. PRISM architecture. The upper row generates and diagnoses the original prediction. The lower row performs historical retrieval and independent verification before the controller selects acceptance, correction, or abstention.

E. Verification

Before a proposed correction is accepted, PRISM checks whether it is consistent with the available evidence. This is important because similar historical examples do not necessarily mean that the alternative prediction is correct. The verification stage evaluates the original prediction, the proposed alternative and the retrieved neighbours:

$$v_i = V_\eta\left(x_i, \hat{y}_i, \hat{y}_{c,i}, \mathcal{N}_K(z_i)\right). \tag{10}$$

The resulting score v_i indicates the support level of the candidate correction. A correction is considered only when it exceeds the required verification threshold. The verification rule and its threshold are selected using validation data and kept fixed during testing.

F. Decision and Abstention

The final stage combines the estimated error risk with the verification result to determine what should happen to the original prediction. PRISM supports three possible outcomes: accepting the original prediction, replacing it with a verified correction or abstaining when neither decision is sufficiently supported.

$$d_i = \begin{cases} \text{ACCEPT}, & r_i \le \tau_r, \\ \text{CORRECT}, & r_i > \tau_r \wedge v_i \ge \tau_v, \\ \text{ABSTAIN}, & \text{otherwise.} \end{cases} \tag{11}$$

In this way, PRISM does not force a correction whenever the detector identifies a potentially unreliable prediction.

V. Experimental Setup

A. Dataset and Prediction

The experiments use ball by ball IPL data obtained from Cricsheet [5]. The objective is to predict the outcome of a match during its 2nd innings. For this, the delivery level records are converted into over level data, each representing the match situation at a particular stage of the chase. Only information available up to that point is used to construct each data.

Algorithm 1 PRISM inference procedure

Require: State x_i, base model f_θ, memory $\mathcal{M}$
Ensure: Decision d_i and optional final prediction $\tilde{y}_i$

$\hat{p}_i \leftarrow f_\theta(x_i)$
$\hat{y}_i \leftarrow \mathbb{I}[\hat{p}_i \ge 0.5]$
$\phi_i \leftarrow \text{ExtractSignals}(x_i, \hat{p}_i)$
$r_i \leftarrow g_\psi(\phi_i)$
if $r_i \le \tau_r$ **then**
  return $(\text{ACCEPT}, \hat{y}_i)$
end if
$z_i \leftarrow h_\omega(x_i, \hat{p}_i, r_i)$
$\mathcal{N}_K \leftarrow \text{Retrieve}(z_i, \mathcal{M}, K)$
$\hat{y}_{c,i} \leftarrow C_\omega(x_i, \hat{y}_i, \mathcal{N}_K)$
$v_i \leftarrow V_\eta(x_i, \hat{y}_i, \hat{y}_{c,i}, \mathcal{N}_K)$
if $v_i \ge \tau_v$ **then**
  return $(\text{CORRECT}, \hat{y}_{c,i})$
else
  return $(\text{ABSTAIN}, \perp)$
end if

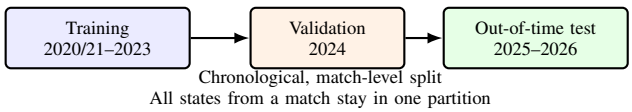


Fig. 2. Documented chronological evaluation protocol for the IPL case study.

B. Chronological Partitioning

To reflect how the model would be used in practice matches are divided chronologically rather than randomly:

- Training: IPL seasons 2020/2021–2023.
- Validation: IPL 2024.
- Testing: IPL 2025–2026.

The partitioning strategy is shown in Fig. 2. All observations belonging to a particular match remain in the same partition, preventing information from a match from appearing across training and testing.

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C. Baseline Models

We compare PRISM against three commonly used classifiers: Logistic Regression (LR), Random Forest (RF) and HistGradientBoosting (HGB). These models provide three different approaches to classification: a linear probabilistic model, a bagging based tree ensemble and a gradient boosted tree model, respectively [6]–[8]. Their preprocessing procedures, feature sets, hyperparameters and random seeds are reported to make the experiments reproducible.

D. Machine Learning Methodology

The prediction task is treated as binary classification at each stage of the 2nd innings. Let $x_{m,o} \in \mathbb{R}^d$ represent the feature vector for match m after over o and let $y_m \in \{0, 1\}$ indicate whether the chasing team eventually wins. Each feature vector is constructed using information available up to the corresponding over. Future deliveries, final scores, match identifiers and post match statistics are not included to avoid target leakage.

1) Base Classifiers: Logistic Regression is used as a linear probabilistic baseline. It estimates the probability that the chasing team wins as

$$p_{\mathrm{LR}}(y = 1 \mid x) = \sigma(\beta_0 + \beta^\top x), \qquad \sigma(a) = \frac{1}{1 + e^{-a}}, \tag{12}$$

where the parameters are learned by minimizing binary cross entropy, with L_2 regularization.

Random Forest combines multiple decision trees trained on bootstrap samples and randomized feature subsets. Its final probability estimate is obtained by averaging the predictions of the individual trees:

$$p_{\mathrm{RF}}(y = 1 \mid x) = \frac{1}{T} \sum_{t=1}^{T} p_t(y = 1 \mid x), \tag{13}$$

where T denotes the number of trees [6].

HistGradientBoosting builds trees sequentially with each new tree helping to reduce the errors made by the existing ensemble [7]. For binary classification, it uses logistic loss and converts the final model output into a probability.

All three classifiers are trained using the earlier seasons, selected on the validation data and evaluated on the later seasons. Their hyperparameters, preprocessing settings, random seeds and calibration configurations must be recorded for reproducibility.

2) Error Detector Learning: PRISM uses a second classifier to estimate whether the base model is likely to make an error. For each training observation, the error indicator is defined as

$$e_i = \mathbb{I}[\hat{y}_i \neq y_i]. \tag{14}$$

The detector uses inference time signals ϕ_i, predicted probability, probability margin, uncertainty estimates and similarity to historical states. It's output represents the estimated probability of an incorrect prediction:

$$r_i = P(e_i = 1 \mid \phi_i). \tag{15}$$

A higher value of r_i indicates that the prediction is highly likely to be incorrect. The risk threshold τ_r is selected using validation data according to the chosen risk–coverage objective and is not adjusted using test outcomes.

3) Historical Retrieval and Candidate Generation: When a prediction is flagged, PRISM searches for similar match situations. The current state is first represented by

$$z_i = h_\omega(x_i, \hat{p}_i, r_i). \tag{16}$$

This representation is compared with historical data u_j. A normalized Euclidean distance is used to measure their similarity:

$$d(z_i, u_j) = \sqrt{\sum_{k=1}^{d} \left(\frac{z_{ik} - u_{jk}}{s_k} \right)^2}, \tag{17}$$

where s_k is a feature wise scale calculated from training data. The K closest observations are retrieved with $K = 50$ in the recorded experiments.

The retrieved outcomes are then used to estimate a candidate probability through distance weighted voting:

$$\begin{aligned} p_{\mathrm{KNN}}(y = 1 \mid z_i) &= \frac{\sum_{j \in \mathcal{N}_K(z_i)} w_{ij} y_j}{\sum_{j \in \mathcal{N}_K(z_i)} w_{ij}}, \\ w_{ij} &= \frac{1}{d(z_i, u_j) + \epsilon}. \end{aligned} \tag{18}$$

Here, $\epsilon > 0$ is to avoid division by 0. The candidate label is obtained as

$$\hat{y}_{c,i} = \mathbb{I}[p_{\mathrm{KNN}}(y = 1 \mid z_i) \geq 0.5]. \tag{19}$$

This formulation describes a distance weighted KNN candidate generator. The retrieval memory is restricted to training period observations so that test outcomes cannot influence the correction process [9].

4) Verification and Calibration: A candidate generated from historical neighbours is not accepted automatically. PRISM uses a verification stage to examine whether the proposed alternative has sufficient support in the retrieved evidence. This assessment considers the estimated risk, the strength of the neighbour vote, agreement between retrieved labels, distance statistics and consistency with the original prediction. The resulting verification score is denoted by v_i.

A threshold τ_v determines whether the candidate can be accepted as a correction. This threshold is selected using validation data and remains fixed during testing. Established approaches such as temperature scaling and Platt scaling can be fitted using held out validation predictions [1], [12], [13].

E. Metrics

Accuracy is computed as

$$\mathrm{Acc} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}[\hat{y}_i = y_i]. \tag{20}$$

For a set of accepted predictions $\mathcal{A}$, selective accuracy is

$$\mathrm{Acc}_{\mathrm{sel}} = \frac{\sum_{i \in \mathcal{A}} \mathbb{I}[\tilde{y}_i = y_i]}{|\mathcal{A}|}, \tag{21}$$

TABLE I
BASELINE RESULTS

Model	Accuracy (%)	Brier	ECE
Logistic Regression	65.2	0.218	0.042
Random Forest	68.4	0.198	0.058
HistGradientBoosting	70.2	0.186	0.049

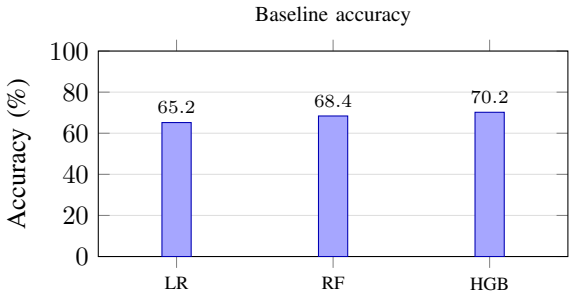


Fig. 3. Baseline accuracy values from the PRISM

and coverage is

$$\mathrm{Cov} = \frac{|\mathcal{A}|}{N}. \tag{22}$$

For binary probabilistic predictions, the Brier score is

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - y_i)^2. \tag{23}$$

Expected calibration error is calculated using B confidence bins, following common calibration practice [1], [13]:

$$\mathrm{ECE} = \sum_{b=1}^{B}\frac{|S_b|}{N}\left|\mathrm{acc}(S_b) - \mathrm{conf}(S_b)\right|. \tag{24}$$

Correction specific metrics should also be reported:

$$\mathrm{EDR} = \frac{\text{base errors detected}}{\text{all base errors}}, \tag{25}$$

$$\mathrm{ERR} = \frac{\text{base errors correctly recovered}}{\text{all base errors}}, \tag{26}$$

$$\mathrm{HCR} = \frac{\text{correct base predictions made incorrect}}{\text{originally correct base predictions}}. \tag{27}$$

VI. RESULTS

A. Baseline Comparison

Table I represents the baseline results available in the PRISM. Their corresponding run level prediction files are required for independent verification. Fig. 3 provides a visual comparison of the 3 baseline models.

B. PRISM Selective Prediction Results

The PRISM results reports 82.4% selective accuracy at 88.5% coverage with Brier score 0.126 and ECE 0.019. The same result reports a harm prevention value of 92.4%.

Selective accuracy and full coverage accuracy measure different quantities. Also the 82.4% selective accuracy must not be directly subtracted from or compared with the baseline accuracies in Table I unless the baseline is re-evaluated on exactly the same accepted subset. A fair comparison should report (i) baseline accuracy over all test states, (ii) baseline accuracy on PRISM's accepted subset, (iii) PRISM selective accuracy on that same subset and (iv) coverage.

TABLE II
PRISM METRICS

Metric	Reported value
Selective accuracy	82.4%
Coverage	88.5%
Brier score	0.126
Calibration error	0.019
Harm prevention	92.4%
Historical states indexed	8,665

TABLE III
SEPARATE 2025 RESULTS II.

Metric	Reported value
Baseline accuracy	80.76%
PRISM selective accuracy	86.21%

C. Ablation Analysis

We conducted ablation experiments to understand how the different PRISM components contribute to the overall performance. Fig. 4 compares the complete pipeline with its modified variants on the 2025 and 2026 test splits. Full PRISM has the highest accuracy in both cases, with a more noticeable performance gap in 2026.

Accuracy alone does not capture the effect of abstention. Fig. 5 compares the coverage and selective risk of the baseline and full PRISM. The baseline makes predictions for all test examples, while PRISM operates at lower coverage with lower selective risk in both years. The recorded coverage is approximately 57% for 2025 and 52% for 2026.

VII. DISCUSSION AND LIMITATIONS

The results are encouraging, but there is still not enough evidence to conclude that PRISM improves prediction quality across all settings. Its performance needs to be considered alongside coverage, since higher selective accuracy may come at the cost of making fewer predictions. In practice, the usefulness of this trade-off depends on how many predictions the application can afford to withhold and how reliable the remaining predictions are. Calibration should also be evaluated separately for all test samples and for the accepted predictions, as these measure different aspects of model reliability.

Finally, the experiments focus only on IPL 2nd innings prediction. Although this provides a useful setting for studying reliability and self-correction, further evaluation on other tasks and datasets is needed to determine whether PRISM can be applied more broadly.

VIII. THREATS TO VALIDITY AND REPRODUCIBILITY

Temporal leakage is a key risk in over level sports prediction. All states from a single match should remain in the same

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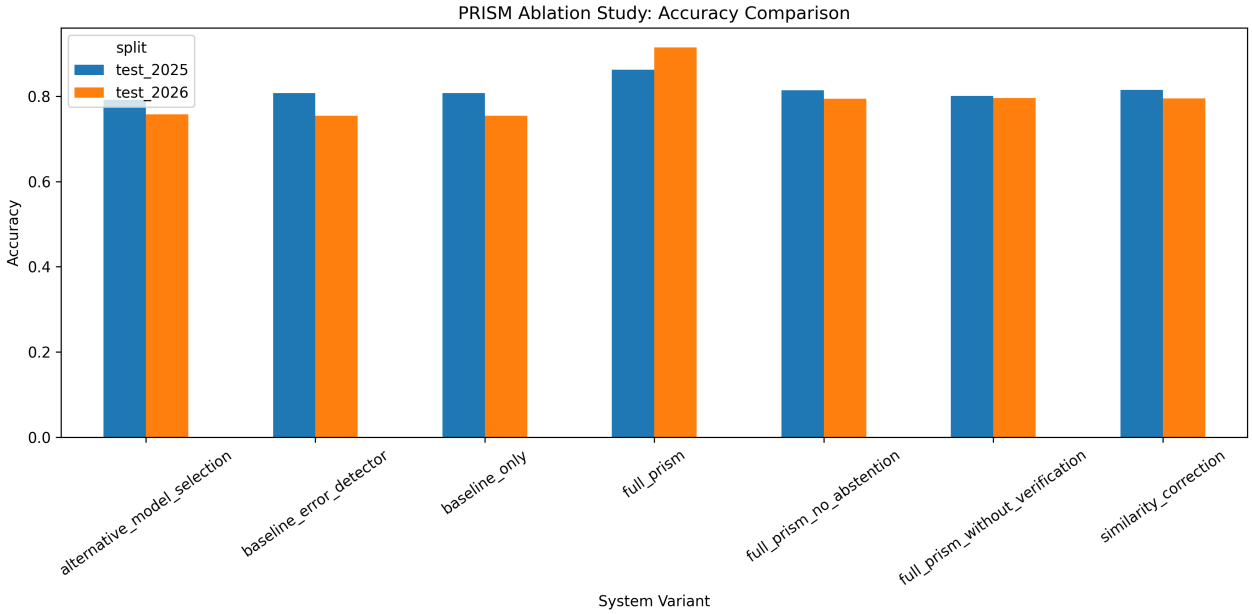


Fig. 4. Accuracy of PRISM ablation variants on the 2025 and 2026 test splits.

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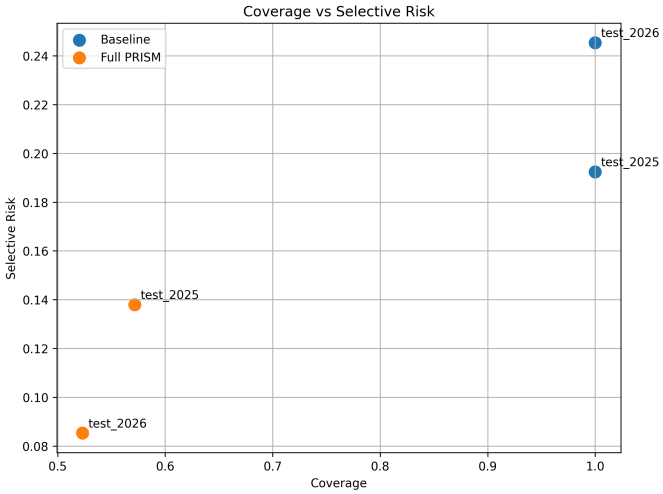


Fig. 5. Coverage and selective risk of the baseline and full PRISM on the 2025 and 2026 test splits.

partition and retrieval at test time must exclude future states and any test labels. Similarly, the error risk detector should be trained using out of fold or held out base model predictions. Thresholds should be selected using validation data only and then frozen before evaluating on the out of time seasons.

IX. Conclusion

This paper presented PRISM, a framework designed to improve the reliability of model predictions after they are generated. It combines error risk estimation, diagnosis, retrieval of similar historical cases and verification of possible corrections. Based on this process, PRISM can retain the original prediction, replace it with a verified alternative or abstain when the available evidence is insufficient.

We evaluated the framework using IPL 2nd innings prediction. In the reported evaluation, PRISM achieved 82.4% selective accuracy at 88.5% coverage with a Brier score of 0.126 and an ECE of 0.019. These results reflect a particular evaluation and should be considered alongside the coverage at which they were obtained.

Future work will focus on more consistent evaluation through matched-subset comparisons, complete ablation reporting and further robustness testing. Testing PRISM on other datasets and prediction tasks will also help establish whether the approach generalises beyond IPL match prediction.

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